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UNDERSTANDING BASIC ELECTRIC THEORY

Introduction

A complete or closed circuit is needed before voltage can cause the current to flow through the load to perform work. A simple electric circuit consists of

1. Voltage source (e.g. battery)
2. Load (e.g. lamp)
3. Path for the current between the power supply and the load (e.g. wire).

The figure below shows a simple electric circuit with its schematic representation. Also it can be seen from the figure (1.0) the current flows in the circuit.

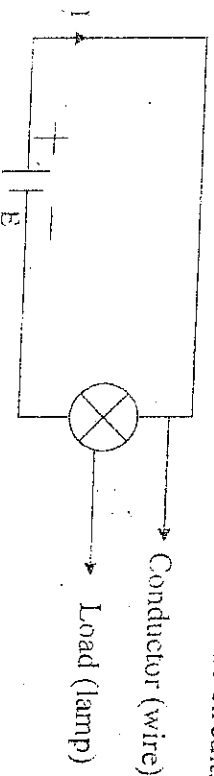


Fig (1) simple diagram of electrical circuit

Normally the current flows from the positive terminal of the battery to the load and to the negative terminal.

Basic Concepts

The three main concepts in electricity are voltage, current and resistance. Voltage or potential difference can be thought as the driving force (although it is not really a "force") behind the electric current.

In the water and current analogy, in figure (2) you can think of voltage as the pressure difference created by a pump that causes the water to flow through the pipe in the water system. Also you can think of the electrons flow in the electric circuit as equivalent to the flow of the water in the pipes that caused by the pump.

Another important concept in electricity is resistance. It's Property of a material that opposes the flow of electrons through it. Resistance in a material is a result of the collision of electrons with the atoms and with each other as they move.

The collisions produce heat, increasing the temperature of the material. This can be useful in some application such as the filament lamp.

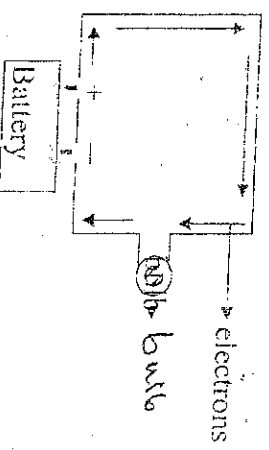
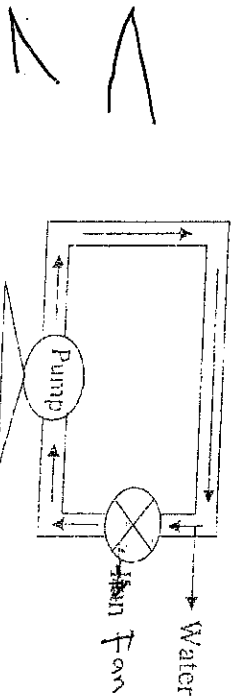


Figure (2) the analogy between water flow and current flow

Note that; bulb, fan are the load while battery and pump are the power source.

The relationship between voltage, current, and resistance in an electrical circuit is fundamental to the operation of any circuit or device.

It can be related mathematical in a law called Ohms law: $V = I \times R$

Capacitors

Voltage (V); is the electrical "pressure" that causes free electrons to flow through an electrical circuit. Its measures in Volt (V)

Current (I); is the Flow of electrons. Its measures in Ampere (A)

Resistance (R); is the Opposition of the materials to the flow of the electric current through it. It measures Ohm (Ω).

To make the previous circuit more practical, a switch and a simple fuse can be added as it can be seen in some circuits. The purpose of the switch is to open or close the circuit i.e. to control the flow of current in the circuit. Note that when the circuit is close the light is on, and when the circuit is open the light is off. the purpose of the fuse in any is to open the circuit when the current exceeds the limit value i.e. for protection.

Voltage Source

Voltage sources can cause two types of currents to flow in the circuit they are;

- (i) Direct Current (DC) and
- (ii) Alternating current (AC).

In DC current, the current flow in one direction only, Example of DC voltage source is batteries and photocells. And in the AC current, the current flows in two directions, i.e. both the positive and the negative half cycle.

Examples of AC current are AC generators and socket outlets. The figure below shows the wave form of AC and DC respectively.

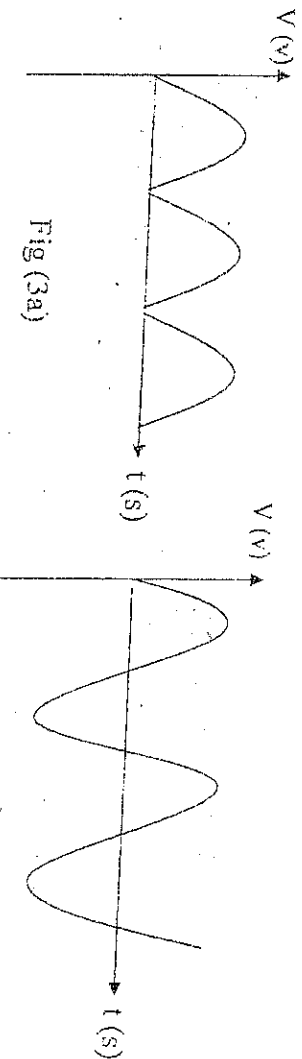


Fig (3a)

Fig (3b)

THE LOAD; Figure: (3a) DC wave forms and

(3b) AC waveforms

The load is what makes use of the energy or power produces by the voltage source. The load can be a lamp, resistor, fan or any domestic appliances. Remember the load can be resistance which produces heating effect (e.g) electric iron, gas cooker etc

THE PATH;

The path is a gateway for the current supply by the voltage source to follow there by reaching the load. The path is usually a good conductor of electricity such as aluminum wire, cooper wire etc, Note that for a current to flow and reach the load, the circuit must be closed by a switch or a key.

OHMS LAW

Ohms law States that the current that is passing through a metallic conductor is directly proportional to the potential difference across its end provided that the temperature and other physical property remain constant.

Mathematically, $V \propto I$

$$V = IR \quad \text{in volt (V)}$$

(i)

$$I = V/R \quad \text{in Amp (A)}$$

(ii)

$$R = V/I \quad \text{in Ohms } (\Omega)$$

(iii)

EXPLANATION OF OHM'S LAW

In the electric circuit, if the voltage across constant resistor value is increase, the current through the resistor will also increase. And if the voltage is decrease, the current will also decrease.

For example, if voltage is double, the current will be double. If the voltage is halved, the current will also be halved. This relationship is illustrated in the figure (4) which used voltage and current meters.

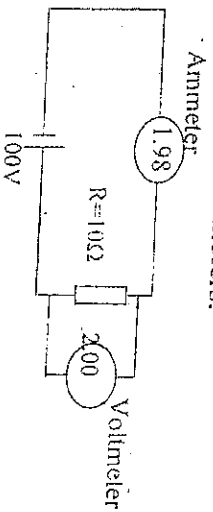


Fig (4a) Voltage decrease, current decrease

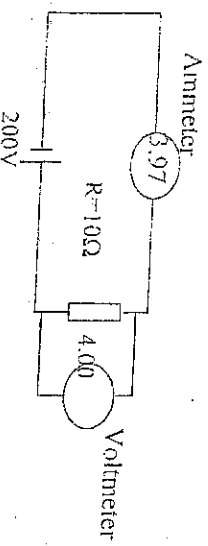


Fig (4b) Voltage increase, current increase

(e.g) calculate the value of the current flow in a circuit when 10Ω resistor and $100V$ is applied.

Solution; given $R = 10\Omega$, $V = 100V$ and $I = ?$

$$\text{Using } I = V / R = 100 / 10 = 10A$$

$$[\text{Therefore } I = 10A]$$

KIRCHHOFF'S LAW

- (i) Kirchhoff's voltage law (KVL) states that the algebraic sum of the product of current and resistances in each of the conductor in any closed path or mesh in a network plus the algebraic sum of the e.m.f in that path is equal to zero.
- Mathematically $\sum IR + \sum e.m.f = 0$
- Consider the network below

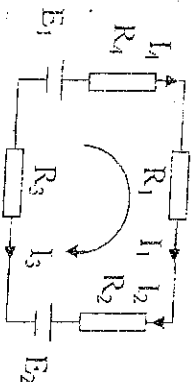


Fig (5a)

Applying Kirchhoff's law to the mesh ABCDA in the fig (5a)

$$-I_1 R_1 - I_2 R_2 + I_3 R_3 - I_4 R_4 - E_2 - E_1$$

$$I_1 R_1 + I_2 R_2 - I_3 R_3 + I_4 R_4 = E_1 - E_2$$

(ii)

Kirchhoff's current law (KCL) states that in any electrical network, the algebraic sum of current meeting at a point or junction is equal to zero.

Mathematically; $\sum I = 0$

Consider the case of a few conductors meeting at a point A in figure (5b) below; some conductors have current leading to point A, where as some have current leading away from point A.

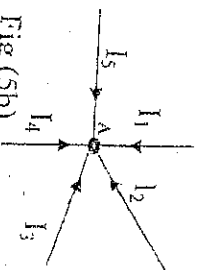


Fig (5b)

Assuming the incoming currents to be positive and outgoing current to be negative i.e

$$I_1 + (-I_2) + I_3 + (-I_4) + I_5 = 0$$

$$I_1 - I_2 + I_3 - I_4 + I_5 = 0$$

Or incoming current = outgoing current.

$$I_1 + I_3 + I_5 = I_2 + I_4$$

THEVENIN'S THEOREM

Thevenin's theorems state that a linear two terminal network may be reduced to an equivalent circuit consisting of a single voltage source in series with a single resistor in resulting open circuit voltage will be the value of the Thevenin's voltage, E_{th} . As shown in figure (6).

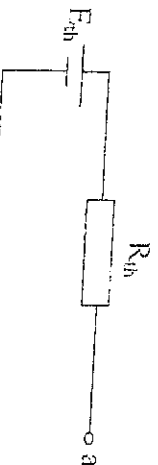


Fig (6) Thevenin's equivalent circuit

BASIC PRINCIPLES OF THEVENIN'S THEOREM

The following principles provide a technique that converts any circuit into its Thevenin's equivalent:

- (1) Remove the load from the circuit.
- (2) Label the resulting two terminals. We will label them as a and b, although any notation may be used.
- (3) Set all sources in the circuit to zero.

Voltage source are set to zero by replacing them with short circuits.

Current sources are set to zero by replacing them with open circuits.

- (4) Determine the Thevenin's equivalent resistance, R_{th} , by calculating the resistance "seen" between terminals, a and b. It may be necessary to redraw the circuit to simplify this step.

- (5) Replace the sources removed in (3), and determine the open circuit voltage between the terminals. The resulting open circuit voltage will be the value of the Thevenin's voltage, E_{th}

- (6) Draw the Thevenin's equivalent circuit using the resistance determined in (4) and the voltage calculated in (5). As part of the resulting circuit, include that portion of the network removed in (1).

ELECTRONIC COMPONENTS

There are two classes of electronic components

- (i) Active component
- (ii) Passive component

✓

Both these electronic component are different from each other

Active electronic components: active electronic components are those that control the flow of electricity. Most electronic printed circuit boards have at least one active component. Some examples of active component are transistors, vacuum tubes, silicon controlled rectifiers (SCRs) etc

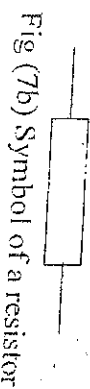
Passive electronic components; Passive electronic components are those that do not have the ability to control current by means of another electrical signal. Examples of passive components are capacitors, resistors, inductors, transformers, diodes etc

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- (i) **Resistor:** a resistor is an electrical device that resists the flow of electrical current. It is a passive device used to control or impede the flow of electric current in an electric circuit by providing resistance, thereby providing a drop in voltage across the device.



Fig (7a) picture of a resistor



- Types of resistor:
- There are two classes of resistors
- (a) Fixed resistors
 - (b) Variable resistors

Fixed Resistor

Fixed resistors are cheap circuit elements which normally do not produce problems. But they represent the largest group of circuit elements in electronic circuits and therefore deserve some consideration. To understand the particular properties of different types of resistors it is necessary to have a look at their construction.

Variable Resistor

Manually variable resistors are used whenever changed during operation or alignment of equipment. The resistance is varied by changing the length of the resistive path between 0 and maximum. Variable resistors are always denominated by their maximum resistance. They normally have three terminals (beginning, end, and slider), so they can be used as resistor or voltage divider.

The majority of the variable resistors are rotating types with circular (approximately 270°) resistive tracks. For special applications linear variable resistors are available. Potentiometer or finders we call the variable resistors which are manipulated during operation. Variable resistors are produced using the same technologies as for fixed resistors. But their construction always requires some more or less complex mechanics as for fixed resistors. subject to wear. Variable resistors are considered relatively unreliable electronic devices.



Fig (8) symbol of variable resistor

Uses of a Resistor

- (i) Resistors are used with transducers to make sensor sub-system (e.g.) fire alarm circuit, smoke sensor alarm circuit etc. this is achieved with light dependent resistor (LDR).
- (ii) Resistors are used to determine the voltage gain of an amplifier.
- (iii) Resistors are also used with capacitors to introduce time delay.
- (iv) Resistors (rheostat) are often used to vary current in a circuit (e.g.) to control the brightness of the lamp.
- (v) In other circuits, resistors are used to direct current flow to particular parts of the circuit.

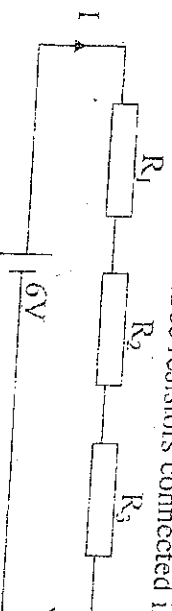
Most electronic circuits require resistors to make them work properly and it is obviously important to find out something about the different types of resistor available, and to be able to choose the correct resistor value, in Ω , $K\Omega$ or $M\Omega$ for a particular application.

Resistors arrangements

(i)

Resistors in series: in a series circuit, the current flowing is the same at all points.

The circuit diagram shows three resistors connected in series with a 6V battery.



It doesn't matter where in the circuit the current is measured, the result will be the same. The total resistance is given by:

$$R_T = R_1 + R_2 + R_3 + \dots + R_n$$

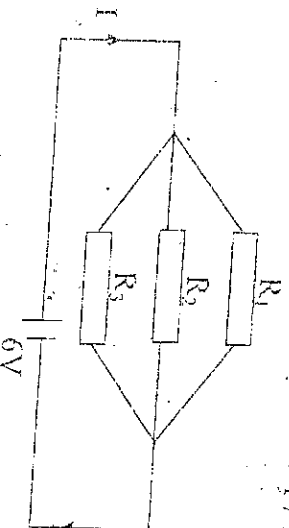
In the circuit above, $R_T = R_1 + R_2 + R_3$

$$R_T = 1 + 1 + 1 = 3K\Omega.$$

(ii)

Resistors in parallel: The Parallel circuits the potential difference across the conductor is the same but different current flows through them. Parallel circuits always provide alternative pathways for current flow. The circuit below shows three resistors connected in parallel to a 6V battery. The total resistance in parallel is given as,

$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3 + \dots + 1/R_n$$



Circuit with series and parallel resistors
Consider the circuit in figure (9) below.

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$$P = VI$$

$$V \frac{V}{V} = \frac{V}{R}$$

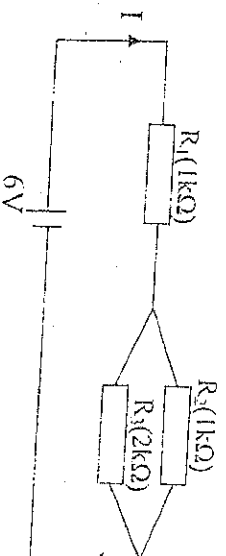


Figure (9) resistor in series and parallel connection

To find the overall resistance, the first step is to calculate the resistance of the parallel connection then adds with the series. From the circuit above

$$R_T = R_1 + R_2 // R_3$$

$$R_T = R_1 + R_2 R_3 / R_2 + R_3$$

$$R_T = 1 + \{ (1 \times 2) / (1 + 2) \} = 1 + 2 / 3 = 1.67 K\Omega$$

Therefore to find the value of current I,

$$\text{Using } I = V / R = 6V / 1.67 K\Omega = 3.59 \text{mA}$$

RESISTOR COLOUR CODE

Resistor colour code or coding help an individual to know or determine the value of the resistor without the use of a meter. The table below shows the various resistor colour code and their values.

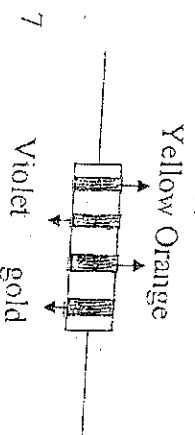
Resistor colour	value
Black	0
Brown	1
Red	2
Orange	3
Yellow	4
Green	5
Blue	6
Violet	7
Grey	8
White	9

Table (1) resistor and colour code and values

The fourth band is the tolerance band i.e the percentage of error. It usually comes in two colour, the gold indicates $\pm 5\%$ and the silver indicates $\pm 10\%$.

(e.g) determine the value of the resistor in figure below?

Solution: Yellow, Y = 4
Violet, V = 7
Orange, O = 3



Therefore $47 \times 10^3 = 47K\Omega$

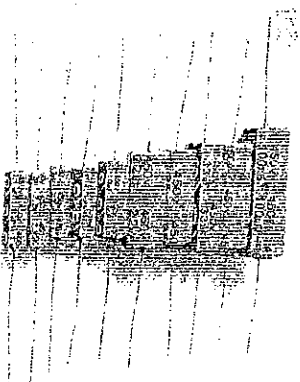
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electron

[Signature]

(ii) **Capacitor:** a capacitor is a passive ~~electrical~~ component that can store energy in the electric field between a pair of conductors called 'plates'. The process of storing energy in the capacitor is known as charging. The ability of a capacitor to store charge is measured by its capacitance.

A wide variety of capacitors are available including electrolytic capacitor, basic parallel plate capacitors and mechanical variable capacitors



Capacitance

Fig (10a) capacitors

This is a measure of a capacitor's and its ability to store charge. A large capacitance means that more charge can be stored. Capacitance is measured in farads, symbol (F).

Capacitor types:-

Capacitors are classified in two namely, polarized capacitor and un-polarized capacitor.

- (a) Electrolytic Capacitors (polarized with large values, μF and above)
- (b) Tantalum Bead Capacitors (polarized)
- (c) Un-polarized capacitors (small values, up to $1\mu F$).

Electrolytic capacitors; are polarized and they must be connected the correct way round, At least one of their leads will be marked + or - if they are not damaged by heat when soldering.

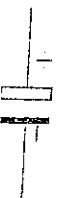


Fig (11) symbol of electrolyte capacitor

Tantalum bead capacitors; are polarized and have low voltage ratings like electrolytic capacitors. They are expensive but very small, so they are used where a large capacitance is needed in a small size.

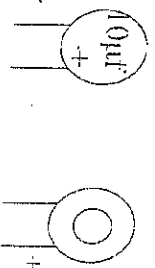


Fig (12) physical symbol of tantalum bead capacitor

Un-polarized capacitors (small values, up to 1 μ F): are Small value capacitors are un-polarized and may be connected either round. They are not damaged by heat when soldering, except for one unusual type. They have high voltage ratings of at least 50V; usually 250V or so. It can be difficult to find the values of this small capacitor because there are many types of them and several different labeling systems.

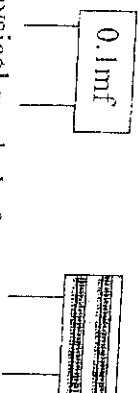
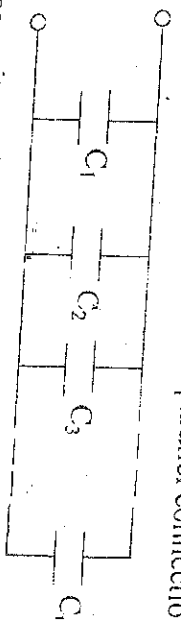


Fig (13) physical symbol of un-polarized capacitor

Capacitor connections:

(i) Parallel connections:

Capacitors in a parallel configuration each have the same potential difference (voltage). The reason for putting capacitors in parallel is to increase the total amount of charge stored. In other words, increasing the capacitance will also increase the amount of energy that can be stored. Figure below is the parallel connection of capacitor.

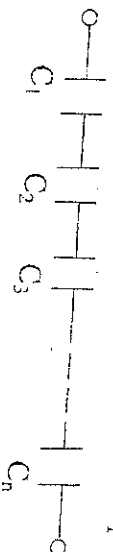


The total capacitance in parallel (C_T) is given by:

(ii) Series connection:

$$C_T = C_1 + C_2 + C_3 + \dots + C_n$$

The current through capacitors in series stays the same, but the voltage across each capacitor can be different. The sum of the potential differences (voltage) is equal to the total voltage. The reason for putting capacitors in series is that we get less capacitance and less charge storage than with either alone (the total voltage is divided between the number of capacitors). Figure below is the series connection of capacitor.



In parallel, the total charge stored is the sum of the charge in each capacitor. While in series, the charge on each capacitor is the same.

The total capacitance in series $1/C$ is given by:

$$1/C_T = 1/C_1 + 1/C_2 + 1/C_3 + \dots + 1/C_n$$

(e.g) two capacitors of capacitance 3 μ F and 6 μ F are connected in series. Calculate the equivalent capacitance?

Solution; given $C_1 = 3\mu$ F, and $C_2 = 6\mu$ F

$$\text{Using } 1/C = 1/C_1 + 1/C_2 = 1/3 + 1/6$$

$$C = \{ (3 \times 6) / (3 + 6) \} = 18 / 9 = 2\mu\text{F}$$

$$\text{Therefore } C = 2\mu\text{F}$$

(e.g) if two capacitors of capacitance $3\mu\text{F}$ and $4\mu\text{F}$ are connected in parallel and another capacitor of capacitance $5\mu\text{F}$ is connected series find the capacitance of a capacitors in a circuit

Solution; Using $C_T = C_1 + C_2 + 1 / C_3$

$$C_T = 3 + 4 + 1 / 5 = 7 + 1 / 5$$

$$C_T = 35 + 1 / 5 = 36 / 5 = 7.2\mu\text{F}$$

Uses of a Capacitor

- (i) Capacitors are used in electronic circuits as energy storage devices.
- (ii) They are used to differentiate between high and low frequency signals.
- (iii) Capacitors can be used with resistors in timing circuits, because it takes time for a capacitor to fill with charge.
- (iv) They are used to smooth varying DC supplies by acting as a reservoir of charge.
- (v) They are also used in filter circuits, because capacitors easily pass AC (changing) signals but they block DC (constant) signals.

- (iii) **Diode:** a diode is a one way valve offor electricity. Diode allows flow of current or electricity in one direction only. Most diode has a painted line on one end showing the direction of flow. The negative side is normally white.

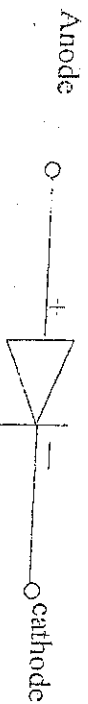


Fig (14a) symbol of diode



Fig (14b) diode

Types of Diode

- There are basically three types of diode namely
- (a) Zener diode.
 - (b) Light emitting diode.
 - (c) Photo diode.

Zener diodes; are used to, provide voltage reference. They also have the nominal voltage rating between 5V and 200V. It's a special diode which is used to maintain a fixed voltage across its terminals.

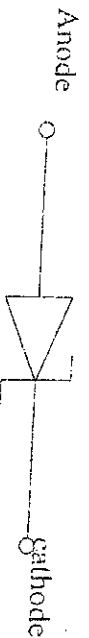


Fig (15) symbol of Zener diode

Light emitting diode (LED); this is a diode which acts as a transducer that converts electrical energy to light.

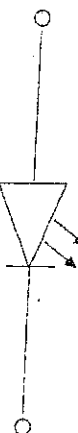


Fig (16) symbol of LED

Photo diode; this is a light sensitive diode.

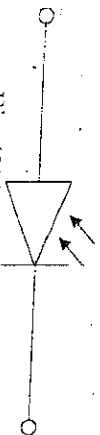


Fig (17) symbol of photo diode.

- (iv) **Integrated circuit (IC);** another name for a chip is called, an integrated circuit (IC) is a small electronic device made up of a semi-conductor material. The first integrated was developed in the 1950s by Jack Kilby of Texas Instruments and Robert Noyce of Fairchild semi-conductor.

Integrated circuit (IC) is a thin chip consisting of at least two interconnected semiconductor devices, mainly transistors, as well as passive components like resistors. As of 2004, typical chips are of size 1cm² or smaller, and contain millions of interconnected devices, but larger ones exist as well. ICs categorized into two groups 'digital and linear ICs'

Integrated circuits are packages of several complex circuits. ICs are wide variety of packages and sizes. Their applications are as varied as their packages.

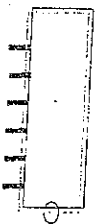


Fig (18a) symbol of IC with 12 pins or legs



Fig (18b) Integrated circuit IC

ICs pin arrangement

- (i) **Single in line:** a single in line (pin) package (SIP or SIPP) has one row of connecting pins. It's not popular as the DIP but has been used for packaging RAM chips and multiples resistors with a common pins. SIPs group RAM chips together on a small board either by DIP process or surface mounting process. The board itself has single row of pins leads that reassemble a component extending from its bottom edge, which plug into a special socket on a system or system expansion board. SIPs are commonly found in memory modules.



Fig (19) single in line pin arrangements

- (ii) **Dual in line:** in microelectronics a dual in line package (DIL) or dual in line pin package (DIPP) is an electronic component package with a rectangular housing and two parallel rows of connecting pins. The package may be through hole soldering and to the printed circuit board or inserted in a socket. Sockets allow easy replacement of a device and eliminate the risk of damage from overheating during soldering. The dual in line format was invented by Don Forbes, Rex Rice and Bryant at Fairchild R&D in 1964, when the restricted number of leads available on circular

Packages become a limitation in the use of integrated circuits.



Fig (20a) dual in line pin arrangements

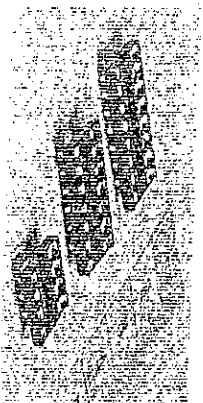


Fig (20b) dual in line socket pin arrangements

- (iii) **Quad in line:** Rockwell used a quad in-line package with 42 leads formed into staggered rows for their PPS-4 microprocessor family introduced in 1973, and other microprocessors and microcontrollers, some with higher lead counts, through the early 1990s.

The QIP, sometimes called a QIL package, has the same dimensions as a DIL package, but the leads on each side are bent into an alternating zigzag configuration so as to fit four lines of solder pads (instead of two with a DIL). The QIL design increased the spacing between solder pads without increasing package size, for two reasons:

(a) First it allowed more reliable soldering. This may seem odd today, given the far closer solder pad spacing in use now, but in the 1970s, the heyday of the QIL, bridging of neighboring solder pads on DIL chips was an issue at times,

(b) QIL also increased the possibility of running a copper track between 2 solder pads. This was very handy on the then standard single sided single layer PCBs. Some QIL packaged ICs had added heat sinking tabs, such as the HAI306.

Intel and 3M developed the ceramic leadless quad in-line package (QLIP), introduced in 1979, to boost microprocessor density and economy. The ceramic leadless QLIP is not designed for surface-mount use, and requires a socket. It was used by Intel for the iAPX 432 microprocessor chip set and by Zilog for the Z8-02 external-ROM prototyping version of the Z8 microcontroller

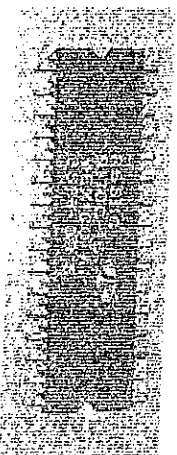


Fig (21) quad in line pin arrangements

- (v) **Transistor:** A transistor is a semi-conductor device. It is the fundamental building block of the circuitry in mobile phone, computers and several electronic devices. A transistor has very fast response.

The transistor is a solid state semiconductor device which can be used for amplification, switching, voltage stabilization, signal modulation and many other functions. It acts as a variable valve which, based on its input current (BJT) or in voltage (FET), allows a precise amount of current to flow through it from the circuit's voltage supply.

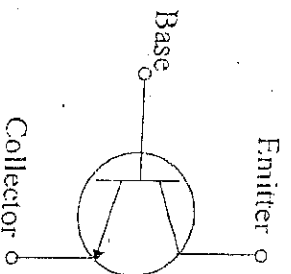
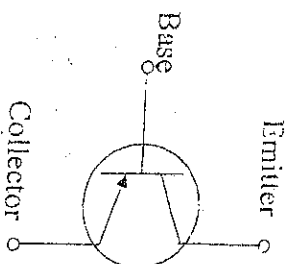


Fig 22(a) Symbol of NPN transistor



(b) symbol of PNP transistor

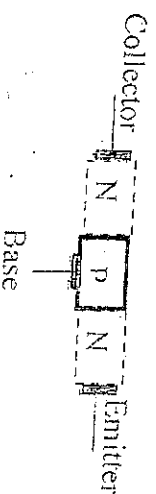
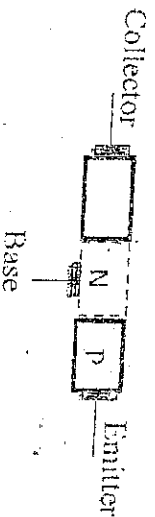
Transistors may be packaged individually or they can be a part of an integrated circuit. Some of the ICs have billions of transistors in a very small area.

Types of Transistor

- (i) Bipolar junction transistor (BJT)
- (ii) Field effect transistors (FET)

A **bipolar junction transistor (BJT)** is a type of transistor, an amplifying or switching device constructed of doped semiconductors that employs *both* types of charge carriers: electrons and holes. The BJT is a three layer sandwich of different types of charge sections, either N-type |P-type |N-type (NPN transistors) or P-type |N-type |P-type (PNP transistors). The center layer is called the *base* of the transistor and is made from lightly doped, highresistivity material. By varying the current into the base terminal, the current allowed to flow between the *emitter* and a third terminal known as the *collector* (which are both heavily doped and hence low resistivity regions) can be varied.

This effect can be used to amplify the input current. BJTs can be thought of as current-controlled current sources and are usually characterized as current amplifiers. Early transistors were made from Typical Transistors germanium but most modern BJTs are made from silicon.



Junction of transistors

A junction of a transistor consists of a thin piece of one type of semiconductor material between two thicker layers of the opposite type. For example, if the middle layer is P-type the other side must be n-type. Such a transistor is an NPN transistor. One of the outside layers is called the *emitter*, and the other is known as the *collector*. The middle layer is the places where they *emitter* joins the base and the base joins the collector are called junctions.

Testing a BJT transistor

Transistors can be damaged by heat when soldering or by misuse in a circuit. If you suspect that a transistor may be damaged there are two easy ways to test it.

Testing Transistor with a multimeter

Use a multimeter or a simple tester (battery, resistor and LED) to check each pair of leads for conduction. Set a digital multimeter to diode test and an analogue multimeter to a low resistance range. Test each pair of leads both ways (six tests in total):

- (i) The base-emitter (BE) junction should behave like a diode and conduct one way only.
- (ii) The base-collector (BC) junction should behave like a diode and conduct one way only.
- (iii) The collector-emitter (CE) should not conduct either way.

The diagram shows how the junctions behave in an NPN transistor. The diodes are reversed in a PNP transistor but the same test procedure can be used.

Field Effect Transistors; these are transistor that has very high input impedance and it has three terminal called; drain, gate and source. A field effect transistor has only two layers of semiconductor material, one on top of the other. Electricity flows through one of the layers, called the channel. A voltage connected to the other layer, called the gate, interferes with the current flowing in the channel. Thus, the voltage connected to the gate controls the strength of the current in the channel.

There are two basic varieties of field effect transistors;

- (i) The junction field effect transistor (JFET) and
- (ii) The metal oxide semiconductor field effect transistor (MOSFET). Most of the transistors contained in today's integrated circuits are MOSFETs.

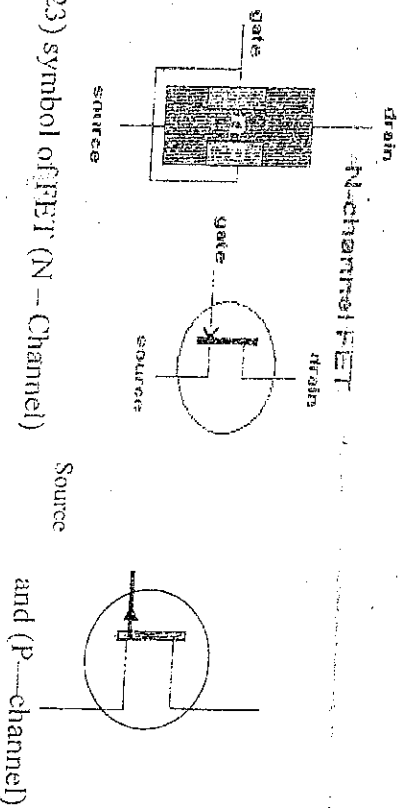


Fig (23) symbol of FET (N - Channel)

Uses of transistor

- (i) Are used in voltage regulation.
- (ii) Amplification.
- (iii) Switching.
- (iv) Signal modulation.
- (v) Oscillators.

POWER RATING

When current flows through a resistance, electrical energy is converted into heat. This is obvious in an electric torch where the lamp filament heats up and glows white hot.

Although the result may be less evident or imperceptible, exactly the same process of energy conversion goes on when current flows through *any* electronic component. The power output of a lamp, resistor, or other component, is defined as the rate of change of electrical energy to heat, light, or some other form of energy. Power is measured in watts, W, or mW, MW, and can be calculated from:

$$P = VI \dots\dots\dots (viii)$$

$$\text{Also } P = V^2/R \dots\dots\dots (ix)$$

$$\text{Also } P = I^2 R \dots\dots\dots (x)$$

Where P is power.

(e.g) what is the power output of a resistor when the voltage across it is 6V, and the current flowing through it is 100mA?

Solution; Using, $P = VI$

$$P = 6 \times 100\text{mA} = 600\text{mW} = 0.6\text{W}$$

(e.g) calculate the power dissipated by a heater of 230V and a resistance of 15Ω.

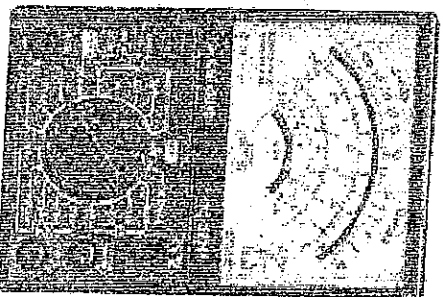
Solution; Using, $P = V^2/R$

$$P = (230)^2/15 = 3526.7\text{W or } 3.526\text{KW}$$

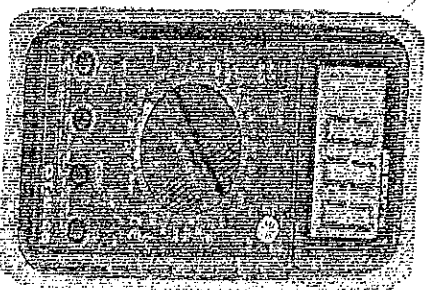
Note; short circuit has zero resistance ($R=0$) while open circuit has infinite resistance ($R=\infty$)

BASIC ELECTRICAL QUANTITIES MEASUREMENT Introduction

It is necessary knowing how to measure voltage, current, and resistance. Special types of instruments are used to measure these basic electrical quantities. The instrument used to measure voltages is a **voltmeter**, the instrument used to measure current is an **ammeter**, and the instrument used to measure resistance is an **ohmmeter**.



(a) Analog multimeter



(b) Digital multimeter

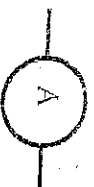
Commonly, all three instruments are combined into a single instrument such as a **multimeter** or **AVO meter** (Ampere-Volt-Ohmmeter), in which you can choose what specific quantity to measure by selecting the switch setting. A shows typical portable multimeter;

Part (a) from figure shows an analog multimeter with pointer, and part (b) from figure shows a digital multimeter with digital screen

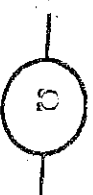
General scheme symbols are used to indicate placement of meters in circuit when value changes need to be shown. Figure (25) shows meter symbols used to present the different meters, as voltmeter, ammeter and ohmmeter.



Voltmeter



ammeter



ohmmeter

Fig (25) shows meter symbols

MEASURING CURRENT WITH AN AMMETER

It knows that current in the circuit is measured by ammeter, to measure the current, the circuit must be open and the ammeter is connected in series of the circuit.

The ammeter connection in the circuit is a series connection.
Figure (26) illustrates how to connect ammeter in the circuit and measure the Current.

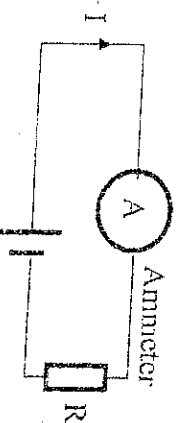
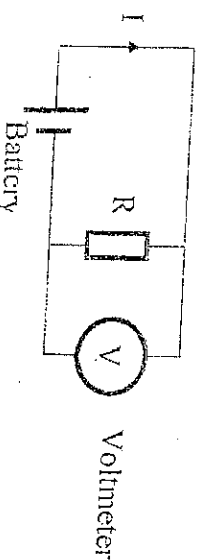


Figure 26: Ammeter

MEASURING VOLTAGE WITH A VOLTMETER

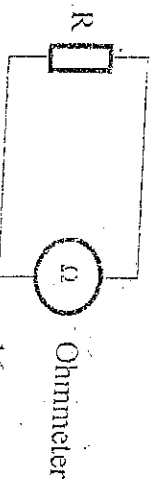
It knows that voltage in the circuit is measured by voltmeter, to measure the voltage, connect the voltmeter across the component which the voltage is to be measured. Voltage can be considered as the pressure that forces the electrons to flow. The voltage is being measured by measuring the difference between the voltages at the two terminals of the device-under test which is the (voltage drop). This can be performed using a measuring instrument called voltmeter.

The voltmeter connection in the circuit is a parallel connection.
Figure 27 below illustrates how to connect voltmeter in the circuit to measure the voltage across the resistor



MEASURING RESISTANCE WITH OHMMETER

To measure resistance, connect the ohmmeter across the resistor. The resistor must first remove from the circuit. This is shown in figure below.



Calculating Current

Example: If the resistance is to $0.1K\Omega$ and the voltage is $50V$, what is the value of current?
Solution: $V = 50$, $R = 0.1K = 100\Omega$, $I = ?$

$$I = V/R = 50/100 = 0.25A \quad 0.25A$$

Calculating voltage

To determine the voltage values when you know the values of resistance and current. As examples, by using the following formula, $V = I R$.

In order to get voltage in volts, you must express the value of current in amperes (A) and the value of resistance in ohms (Ω).

Example: Calculate how much voltage is needed to produce $6A$ of current with $200K\Omega$ resistor
Solution: using $V = I R$

$$V = 6 \times 200 = 1200 V$$

$$V = 1200/1000 = 1.2 KV$$

Example: what is the value of voltage in the circuit if the current is $30mA$ and the resistance is $3.4K\Omega$.

Solution: Using $V = I R$

$$V = 3400 \times 0.03 = 102 V$$

Calculating Resistance

To find the resistance values when you know the values of voltage and current. As examples, by using the formula, $R = V/I$

In order to get resistance by ohms, you must express the value of voltage in volts (V) and the value of current in amperes (A).

Example: how much resistance is needed to draw $4mA$ of current with $24V$ in the circuit?
Solution: Using the formula $R = V / I$

$$R = 24 / 4 = 6\Omega$$

OSCILLOSCOPE;

Is an instrument used to observe or measure the change of an electrical signal over time, such that voltage and time describe a shape which is continuously graphed against a calibrated scale. The observed waveform can be analyzed for such properties as amplitude, frequency, rise time, time interval, distortion and others.

Basics of O-Scopes

The main purpose of an oscilloscope is to graph an electrical signal as it varies over time. Most scopes produce a two-dimensional graph with time on the x-axis and voltage on the y-axis.

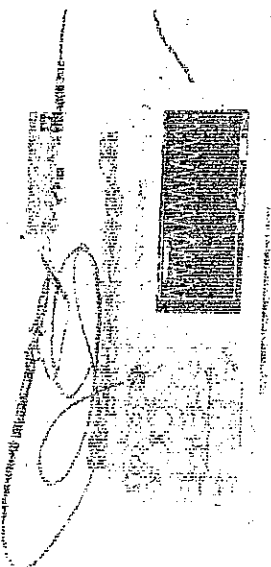


Fig (29) diagram of oscilloscope

Fig (29) an example of an oscilloscope display. A signal (the yellow sine wave in this case) is graphed on a horizontal time axis and a vertical voltage axis.

Controls surrounding the scope's screen allow you to adjust the scale of the graph, both vertically and horizontally – allowing you to zoom in and out on a signal. There are also controls to set the trigger on the scope, which helps focus and stabilize the display.

What Can Scopes Measures

In addition to those fundamental features, many scopes have measurement tools, which help to quickly quantify frequency, amplitude, and other waveform characteristics. In general a scope can measure both time-based and voltage-based characteristics:

(1) Timing-based characteristics:

- (i) Frequency and period – Frequency is defined as the number of times per second a waveform repeats. And the period is the reciprocal of that (number of seconds each repeating waveform takes). The maximum frequency a scope can measure varies, but it's often in the 100's of MHz (1E6 Hz) range.
- (ii) Duty cycle – The percentage of a period that a wave is either positive or negative (there are both positive and negative duty cycles). The duty cycle is a ratio that tells you how long a signal is "on" versus how long it's "off" each period.
- (iii) Rise and fall time – Signals can't instantaneously go from 0V to 5V, they have to smoothly rise. The duration of a wave going from a low point to a high point is called the rise time, and fall time measures the opposite. These characteristics are important when considering how fast a circuit can respond to signals.

(2) Voltage-based characteristics:

- (i) Amplitude – Amplitude is a measure of the magnitude of a signal. There are a variety of amplitude measurements including peak-to-peak amplitude, which measures the absolute difference between a high and low voltage point of a signal. Peak amplitude, on the other hand, only measures how high or low a signal is past 0V.
- (ii) Maximum and minimum voltages – The scope can tell you exactly how high and low the voltage of your signal gets.
- (iii) Mean and average voltages – Oscilloscopes can calculate the average or mean of your signal, and it can also tell you the average of your signal's minimum and maximum voltage.

Uses an Oscilloscope

- The o-scope is useful in a variety of troubleshooting and research situations, including:
- (i) Determining the frequency and amplitude of a signal, which can be critical in debugging a circuit's input, output, or internal systems? So you can say if it's a component in your circuit has malfunctioned.
 - (ii) Identifying how much noise is in your circuit.
 - (iii) Identifying the shape of a wave – sine, square, triangle, saw tooth, complex, etc.
 - (iv) Quantifying phase differences between two different signals.

LOGIC FAMILY

Logic Families indicate the type of logic circuit used in the IC. The main types of logic families are:

- (i) TTL (Transistor Transistor Logic)
- (ii) CMOS (Complementary MOS)
- (iii) ECL (Emitter Coupled Logic)

Characteristics of logic families

The main characteristics of logic families or Digital ICs are complete functioning logic networks. Typically, a Digital IC requires only a power supply, I/P (input) and O/P (output). The characteristics of Logic families or digital ICs include:

- (i) Speed
- (ii) Fan-in
- (iii) Fan-out
- (iv) Noise Immunity
- (v) Power Dissipation

Speed: Speed of a logic circuit is determined by the time between the application of input and change in the output of the circuit.

Fan-in: It determines the number of inputs the logic gate can handle. The fan-in defined as the maximum number of inputs that a logic gate can accept. If number of input exceeds, the output will be undefined or incorrect. It is specified by manufacturer and is provided in the data sheet.

Fan in is a term used to describe the maximum number of inputs which can be given to a logic gate (logic circuit). For example: A 3-input NAND gate has fan in equal to 3.

Fan-out: Determines the number of circuits that a gate can drive. The fan-out is defined as the maximum number of inputs (load) that can be connected to the output of a gate without degrading the normal operation.

Fan Out is calculated from the amount of current available in the output of a gate and the amount of current needed in each input of the connecting gate. It is specified by manufacturer and is provided in the data sheet. Exceeding the specified maximum load may cause a malfunction because the circuit will not be able supply the demanded power.

Noise Immunity: Maximum noise that a circuit can withstand without affecting the output.

Power dissipation: When a circuit switches from one state to the other, power dissipates.

Transistor-transistor Logic (TTL)

In transistor-transistor logic (TTL), logic gates and other digital circuits are made using bipolar junction transistors and resistors. The term transistor-transistor is because both logic function and amplification is done by transistor. Using TTL logic families, many logic gates can be fabricated in a single integrated circuit. For logic gate built using TTL logic families, input are given to the emitters of the input transistor. In TTL logic family, analog value from 0 V to 0.8 V is logic 0 and 2 V to 5 V is logic 1.

Advantages of the TTL logic families include;

- (i) high switching speed (125 MHz),
- (ii) less noise and
- (iii) More current (3 mA) driving capability.

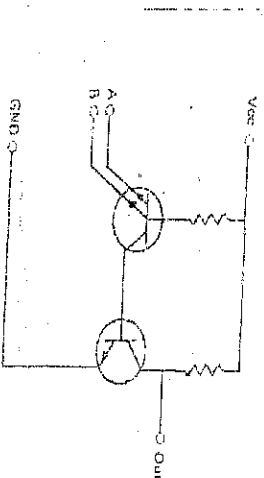


Fig (30) two input NAND gate using TTL

Figure (30) shows TTL NAND gate. There are two transistor stages in the circuit, a multi-emitter input transistor and output transistor. Function of a multi-emitter transistor is same as that of a two parallel transistor with common base and collector terminals.

Operation of TTL NAND Gate

When the two emitters of the input transistors are connected to high voltage, then emitter-base junction of the transistor is reverse biased that means, transistor is in reverse active mode. In reverse active mode, less magnitude current flows in the opposite direction. This current reaches base of the output transistor, allowing it to conduct and pulling down the output voltage to zero. When any one of the input terminal is low, the current through other branch flows out through this terminal. Now no current reaches the base terminal of the output transistor, so output remains at high state.

Emitter Coupled Logic (ECL)

Emitter Coupled Logic (ECL), also referred to as Current Mode Logic families, is a digital technology with extremely high-speed. Transistors are not allowed to go into deep saturation thus, eliminating storage delays like in TTL logic families. Transistors are driven either in cut off or in active region. This is achieved by using voltage values close to each other. For logic one it is -0.9 and for logic zero, it is -1.75v. In the active region, charge stored in the base region of transistors is kept to minimum.

Difference between these two logic states is very small. This improves the speed of operation at the expense of noise margin. Propagation time for an ECL gate is 0.5 to 2ns, which is very less when compared to its TTL counterpart.

The **disadvantage** of ECL logic families is that it uses a negative power supply such that the logic levels are not compatible with any other logic family and makes analysis and measurement inconvenient.

An ECL logic family requires large currents therefore power dissipation is 3 to 10 times higher than that of TTL logic families. Because of its large power consumption and large scale requirement of silicon area, CMOS logic gates are preferred over ECL logic families in large scale integrated circuits.

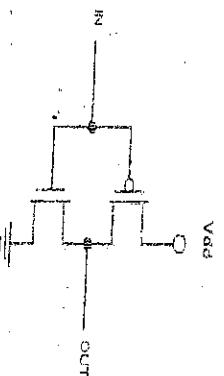


Fig (31) cmos logic

Because of high noise immunity and low static power dissipation, now CMOS logic families is most preferred in large scale integrated circuits. CMOS (Complementary Metal Oxide Semiconductor) has complementary and symmetrical NMOS and PMOS transistors. Figure shown below is a CMOS inverter.

Depending on the input value, only one transistor of the CMOS inverter will be ON at a time. So in both states, there is no direct connection between power supply and ground, thereby reducing static power loss of a transistor.

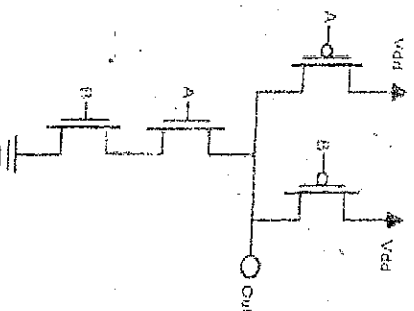


Fig (32) Input CMOS NAND gate two

When both inputs of a CMOS NAND gate are at high, then transistors T1, T2 will be ON and transistors T3, T4 will be OFF. Now there is a direct connection between output and ground through transistor T3 and T4. Output is pulled down to zero.

When any of the input is low, either transistor T3 or T4 will be off. This breaks the connection between output terminal and ground. At the same time, one of the PMOS transistors is ON, so there is connection between output and power supply. Now output is pulled up to logic high.

MAINTENANCE

Maintenance is a combination of any action carried out to retain an item in or restore it to normal operational standard.

There are two major types of maintenance namely;

- (i) Preventive maintenance.
- (ii) Corrective maintenance.

Preventive Maintenance

Preventive maintenance is maintenance in which components or a unit of component of any device (e.g. C.P.U, monitor, keyboard etc) are removed and replaced by new ones at a specific interval of time, even if it's not break down or fails completely in order to improve its efficiency.

Corrective Maintenance

Corrective maintenance is the process of searching for fault in order to correct it or when equipment has failed and repair it back to its normal operation. These normally done when the system breakdown.

Troubleshooting the process of searching for fault in order to correct it, it could be Hardware or software troubleshooting the following is the procedure in locating fault.

Preventative maintenance is maintenance that is regularly performed on a piece of equipment to lessen the likelihood of it failing. Preventative maintenance is performed while the equipment is still working, so that it does not break down unexpectedly.

The act of a regularly scheduled check of the computer hardware or software to help ensure it continues to operate properly. Below is a list of some preventive maintenance steps you can take with your computer or computer hardware. Cleaning your computer hardware. Downloading the latest drivers for your hardware.

Computer Preventive Maintenance

- (i) Cleaning your computer hardware;
- (ii) Downloading the latest drivers for your hardware;
- (iii) Downloading the latest updates for your computer software;
- (iv) Verifying you have the latest anti-virus protection updates on your computer;
- (v) Running disk software utilities such as Defrag and Scan Disk on your hard drive;
- (vi) Deleting unused programs or other files on your computer.

If you don't turn off your computer it may be good to reboot your computer every few months, unless the computer is a critical computer such as a computer server.

IMPORTANCE OF PREVENTIVE MAINTENANCE

Preventive maintenance is one of the most ignored aspects of PC ownership most people seem to think that the PC doesn't need preventive maintenance, and so you should just use it until it breaks, and then repair or replace it. These people generally find themselves repairing or replacing much sooner than those that take definite steps to avoid difficulties in the first place.

The importances of preventive maintenance plan for PC are;

(i) **Preventive Maintenance Saves Money:** I'm sure we've all heard the old adage "an ounce of prevention is worth a pound of cure". It's true, but it is also true. Avoiding problems with your PC will save you money in the long run, compared with laying out cash for new components or repair jobs.

(ii) **Preventive Maintenance Saves Time:** Saves time? How can taking two hours a month or whatever to perform maintenance save time? Simple: because it saves you the much bigger hassles of dealing with system failures and data loss. Most preventive maintenance procedures are quite simple compared to troubleshooting and repair procedures--now those can really eat up your time at a fantastic rate.

(iii) **Preventive Maintenance Helps Safeguard Your Data:** For most people, the data on the hard disk is more important than the hardware that houses it. Taking steps to protect this data therefore makes sense, and that is what PM is all about.

(iv) **Preventive Maintenance Improves Performance:** Some parts of your system will actually degrade in performance over time, and preventive maintenance will help to improve the speed of your system in these respects.

Other importance of preventive maintenance of hardware components are:

- (v) **Maintenance of THE CASE;** Use a clean cloth lightly dampened with ammonia cleaner to remove dust, dirt, or stains from the exterior of the PC. Start at the top and work down. Add a little bit of extra cleaner to remove stubborn stains. You'll find that the housing base is typically the dirtiest (especially for tower systems). When cleaning, be careful not to accidentally alter the CD-ROM volume or sound card master volume controls. Also, do not dislodge any cables or connectors behind the PC.

- (vi) **Maintenance of THE AIR INTAKE;** When cleaning the case, pay particular attention to the air intake(s), which are usually located in the front (or front sides) of the housing check for accumulations of dust or debris around the intakes, or caught in an intake filter. Clean away any accumulations from the intake area then use your static-safe vacuum to clean the intake filter, if possible. You might need to remove the intake filter for better access. If the intake filter is washable, you might choose to rinse the filter in simple soap and water for best cleaning (remember to dry the filter thoroughly before replacing it). Of course, if there is no intake filter, simply clean around the intake area.

- (vii) **Maintenance of THE SPEAKERS;** Multimedia speakers offer a countless number of ridges and openings that are just perfect for accumulating dust and debris. Use a CAN of compressed air to gently dust-out the Always dampens a clean towel with cleaner. Never spray cleaner directly onto any part of the computer. Speaker's openings do not insert the long, thin air nozzle into the speaker you can easily puncture the speaker cone and ruin it. Instead, remove the long nozzle and spray air directly from the can. Afterward, use a clean cloth lightly dampened with ammonia solution to remove any dirt or stains from the speaker housings.

TROUBLESHOOTING METHODOLOGY

Troubleshooting is a method of finding the cause of a problem and correcting it. The ultimate goal of troubleshooting is to get the equipment back into operation. This is a very important job because the entire production operation may depend on the troubleshooter's ability to solve the problem quickly and economically, thus returning the equipment to service. Although the actual steps the troubleshooter uses to achieve the ultimate goal may vary, there are a few general guidelines that should be followed. There are often cases where a familiar piece of equipment or system breaks down. In those cases, an abbreviated five-step troubleshooting process can be used to find the fault, get the system up and running. It is important to note that, although it is a five-step approach, the same basic guidelines of the seven-step troubleshooting method are followed. The steps are simply combined to be specific to the problem at hand. This article will briefly cover the five-step troubleshooting process, followed by a more in-depth look at the seven-step troubleshooting process.

General Troubleshooting Guidelines

The general guidelines for a good troubleshooter to follow are to Use a clear and logical approach;

- (i) Work quickly.
- (ii) Work efficiently.
- (iii) Work economically.

- (iv) Work safely and exercise safety precautions.

Troubleshooting Steps

The five-step troubleshooting process consists of the following:

- (i) Verify that a problem actually exists.
- (ii) Isolate the cause of the problem.
- (iii) Correct the cause of the problem.
- (iv) Verify that the problem has been corrected.
- (v) Follow up to prevent future problems.

Step 1: Verify That a Problem Actually Exists

The troubleshooting process begins with symptom recognition. To troubleshoot, there must first be a problem. In this alternate approach to troubleshooting, the troubleshooter must first verify that there actually is a problem. This is actually a combination of the symptom recognition (is the first step in identifying a failure is recognizing that a failure exists) and symptom elaboration (is the step that you will obtain as much information about the problem as possible i.e. (like what is the problem). To verify that there actually is a problem, the troubleshooter must use all available means of information. This includes the equipment operator, equipment indications and controls, and technical documentation about the equipment or system. Contacting the equipment operator should be the first action taken. The operator usually can supply many of the details concerning the failure incident. To get the most information, the troubleshooter should ask probing questions. Some examples are:

- (i) What are the operators indications of the trouble
- (ii) How did the operator discover the trouble
- (iii) What were the conditions at the time the trouble first occurred
- (iv) Is the trouble constant or intermittent

Step 2: Isolate the Cause of the Problem

The second step of the five-step troubleshooting process relies heavily on the troubleshooter's technical skills and intuition. During this step, the troubleshooter is actively involved in isolating the cause of the problem. This involves physical activity, such as reading instrumentation, connecting test equipment, adjusting parameters, and possible disassembly. It also involves mental activity, such as logic, evaluation, and reasoning. The specialized knowledge of the troubleshooter plays a key part in this step. To safely and effectively isolate the cause of the problem, keep the following in mind:

- (i) Begin investigating the easiest items to check. Eliminate all convenient possibilities first to save time.
- (ii) Be aware of special operating modes (self-tests, built-in diagnostics, etc.) that may aid in the troubleshooting process.
- (iii) Use appropriate safety precautions and equipment when troubleshooting in the field.
- (iv) When taking instruments in a piping system off-line and when placing them on-line, follow appropriate operating procedures.
- (v) Recognize the obvious, but do not ignore what may be concealed.

Using the appropriate documents and test equipment, the troubleshooter continues to eliminate possible causes. As each check is completed, the trouble becomes more isolated.

Use techniques, such as **half-splitting** and **signal tracing**, helps to narrow the problem down quickly. A flowchart also used to illustrating the process used to isolate the cause of the problem. Once the problem has been isolated to a specific component, it can be repaired.

Step 3: Correct the Cause of the Problem

The third step of the five-step troubleshooting process is correcting the cause of the problem. This step involves performance of the repair or other activity that eliminates the problem. This can be as simple as turning a switch or adjusting a valve, or it could be as complex as re-winding a motor or overhauling a pump. To correct the cause of the problem, the troubleshooter performs both failure analysis and a retest of the equipment.

Step 4: Verify That the Problem Has Been Corrected

Once the corrective action is taken, the troubleshooter should verify that the trouble has been corrected. This usually involves rechecking the same indications that proved there was a problem. This time though, the checks should prove that a problem does not exist. This step should be thorough. If there are both an abbreviated procedure and an expanded procedure for checking the equipment, use the expanded procedure. This helps ensure that the problem no longer exists and did not mask another problem. During this verification, the following should be observed:

- (i) Check all indications that relate to the repaired area.
- (ii) Perform a valve/switch line-up check to validate the integrity of the system.
- (iii) Using approved procedures, establish normal operating conditions and check equipment performance.
- (iv) Check for abnormal operation of all inputs and outputs to the repaired equipment.

By thoroughly verifying the proper operation of the repaired equipment, the troubleshooter can be relatively sure the problem has been resolved correctly. To help ensure the problem does not re-occur, the next step in the process is performed.

Step 5: Follow Up to Prevent Future Problems

The final step in the five-step troubleshooting process is to follow up to prevent future problems. This step involves taking preventive measures and recommend actions that could help keep the equipment from failing. This may include the following:

- (i) Changing the preventive maintenance schedule to help prevent failures.
- (ii) Recommend a different supplier if a replacement component is unsatisfactory.
- (iii) Recommend procedure modifications that may prevent future failures.
- (iv) Conduct operator/maintenance training to raise awareness of the potential for problems.
- (v) Complete proper documentation and troubleshooting log entries to aid in future troubleshooting of similar problems.

Although the system retest and preventative measures taken may not seem as vital as fixing the problem and getting the equipment back on-line, these steps are vital to long-term productive performance.

What is Module Testing?

Module testing is a process of testing the individual subprograms, subroutines, classes, or procedures in a program. Instead of testing whole software program at once, module testing recommend testing the smaller building blocks of the program.

The objective of doing Module testing is not to demonstrate proper functioning of the module but to demonstrate the presence of an error in the module. Module testing allows to implement parallelism into the testing process by giving the opportunity to test multiple modules simultaneously.

The advantage of Module testing

- (i) Probability of identifying errors or bugs on smaller chunks of program becomes higher.
- (ii) Multiple modules can be tested simultaneously and hence supports parallel testing.
- (iii) Complexity of testing can be easily managed.

Challenges in module testing (Disadvantages) are

- (i) Non-incremental testing requires more work.
- (ii) Misunderstanding test doubles.
- (iii) Debugging test often.
- (iv) Need to understand the code

For module testing method used is either an **Incremental or non-Incremental method.**

- (i) **Incremental method**- each module is tested first and then gradually incremented to the tested collection. It does a step wise retesting.
- (ii) **Non-incremental method**- all modules are tested independently. First, it combines all modules and then test the whole program.

For Incremental Testing, there are two approaches – Top down and Bottom Up testing.

To execute the module with the selected data, it requires a driver for supplying the test data, monitoring the execution and capturing the results, Module testing is the testing of complete code objects as produced by the compiler when built from source.

A library may be composed of a single compiled object or several compiled objects. There is only a slight difference between unit testing and module testing. Modules are fully formed chunks of coherent source code that can typically be tested by driving a few function signatures with various stimuli. On the other hand, unit testing (which is considered as part of the implementation phase for this software development process) may involve testing one small part of a function that will never formally implement any function interface.

Basic Hardware Components

Hardware is the physical equipment needed for a computer to function properly. The basic hardware parts are briefly described here. Many computer lab managers will already have been exposed to computer hardware through other courses, but those desiring additional information can find it in a variety of texts and online sources. A desktop computer is used in the photos, but all of this equipment is also found (in a more compact arrangement) in a laptop computer.

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(a) **Case.**

The computer case (also called a tower or housing) is the box that encloses many of the parts shown below. It has attachment points, slots and screws that allow these parts to be fitted onto the case. The case is also sometimes called the CPU, since it houses the CPU (central processing unit or processor), but this designation can lead to confusion.

(b) **Power Supply.**

The power supply is used to connect all of the parts of the computer described below to electrical power. It is usually is found at the back of the computer case.

(c) **Fan.**

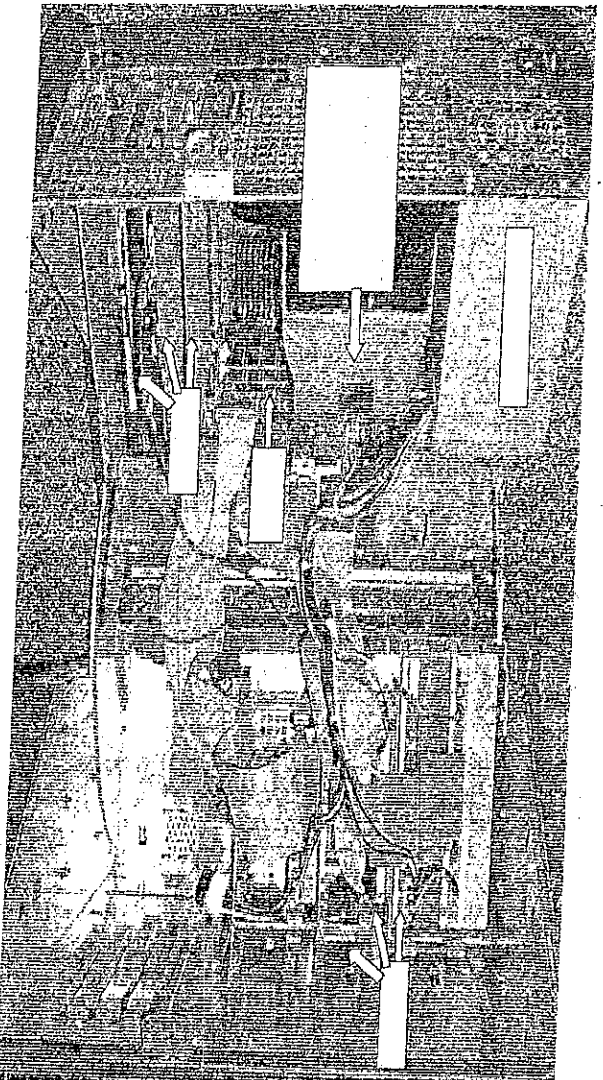
A fan is needed to disperse the significant amount of heat that is generated by the electrically powered parts in a computer. It is important for preventing overheating of the various electronic components. Some computers will also have a heat sink (a piece of fluted metal) located near the processor to absorb heat from the processor.

(d) **Motherboard.**

The motherboard is a large electronic board that is used to connect the power supply to various other electronic parts, and to hold these parts in place on the computer. The computer's memory (RAM, described below) and processor are attached to the motherboard. Also found on the motherboard is the BIOS (Basic Input and Output System) chip that is responsible for some fundamental operations of the computer, such as linking hardware and software. The motherboard also contains a small battery (that looks like a watch battery) and the chips that work with it to store the system time and some other computer settings.

Drives.

A computer's drives are the devices used for long term storage of information. The main storage area for a computer is its internal **hard drive** (also called a hard disk). The computer should also have disk drives for some sort of removable storage media. A **floppy** **disk drive** was very common until recent years, and is still found on many older desk top computers. It was replaced by **CD-ROM** and **DVD** drives, which have higher storage capacities. The current standard is a **DVD-RW** drive, which can both read and write information using both CD and DVD disks. The USB ports (described later) on a computer can also be used to connect other storage devices such as **flash drives** and **external hard drives**.



Cards.

This term is used to describe important tools that allow your computer to connect and communicate with various inputs and output devices. The term "card" is used because these items are relatively flat in order to fit into the slots provided in the computer case. A computer will probably have a sound card, a video card, a network card and a modem. A

RAM.

RAM is the abbreviation for random access memory. This is the short term memory that is used to store documents while they are being processed. The amount of RAM in a computer is one of the factors that affect the speed of a computer. RAM attaches to the motherboard via some specific slots. It is important to have the right type of RAM for a specific computer, as RAM has changed over the years.

Processor.

The processor is the main "brain" of a computer system. It performs all of the instructions and calculations that are needed and manages the flow of information through a computer. It is also called the CPU (central processing unit), although this term can also be used to describe a computer case along with all of the hardware found inside it. Another name for the processor is a computer "chip" although this term can refer to other lesser processors (such as the BIOS). Processors are continually evolving and becoming faster and more powerful. The speed of a processor is measured in megahertz (MHz) or gigahertz (GHz). An older computer might have a processor with a speed of 1000 MHz (equivalent to 1 GHz) or lower, but processors with speeds of over 2 GHz are now common. One processor company, Intel, made a popular series of processors called Pentium. Many reconditioned computers contain Pentium II, Pentium III and Pentium 4 processors, with Pentium 4 being the fastest of these.

Peripheral hardware.

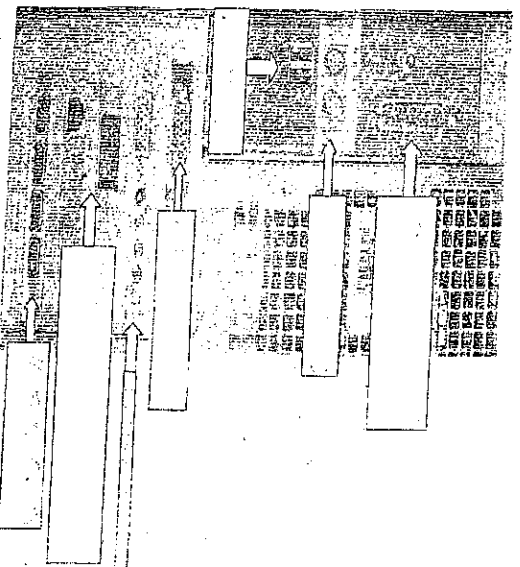
Peripheral hardware is the name for the computer components that are not found within the computer case. This includes input devices such as a mouse, microphone and

keyboard, which carry information from the computer user to the processor, and output devices such as a **monitor**, **printer** and **speakers**, which display or transmit information from the computer back to the user.

Computer Ports

The peripheral hardware mentioned above must attach to the computer so that it can transmit information from the user to the computer (or vice versa). There are a variety of ports present on a computer for these attachments. These ports have gradually changed over time as computers have changed to become faster and easier to work with. Ports also vary with the type of equipment that connects to the ports. A computer lab manager should become familiar with the most common ports (and their uses), as described below.

- (i) **Serial Port.** This port for use with 9 pin connectors is no longer commonly used, but is found on many older computers. It was used for printers, mice, modems and a variety of other digital devices.
- (ii) **Parallel Port.** This long and slender port is also no longer commonly used, but was the most common way of attaching a printer to a computer until the introduction of USB ports (see below). The most common parallel port has holes for 25 pins, but other models were also manufactured.
- (iii) **VGA.** The Video Graphics Array port is found on most computers today and is used to connect video display devices such as monitors and projectors. It has three rows of holes, for a 15 pin connector.
- (iv) **PS/2.** Until recently, this type of port was commonly used to connect keyboards and ~~mouse~~ ^{mouse} to computers. Most desktop computers have two of these round ports for six pin connectors, one for the mouse and one for the keyboard.
- (v) **USB.** The Universal Serial Bus is now the most common type of port on a computer. It was developed in the late 1990s as a way to replace the variety of external storage devices such as DVD-RW drives and flash drives. It has gone through three different models (USB 1.0, USB 2.0 and USB 3.0), with USB 3.0 being the fastest at sending and receiving information. Older USB devices can be used in newer model USB ports.
- (vi) **TRS/TRS** (tip, ring and sleeve) ports are also known as ports for mini-jacks or audio jacks. They are commonly used to connect audio devices such as headphones and microphones to computers.
- (vii) **Ethernet.** This port, which looks like a slightly wider version of a port for a phone jack, is used to network computers via category 5 (CAT5) network cable. Although many computers now connect wirelessly, this port is still the standard for wired networked computers. Some computers also have the narrower port for an actual phone jack. These are used for modem connections over telephone lines.



Basic Hardware Repairs,

Below are descriptions of some of the most common hardware repairs, including key steps in the process. In all cases, however, the lab manager who is just beginning to work on computer repair would be encouraged to complete these repairs alongside a more experienced technician as they gain skill and confidence. If the lab manager has an opportunity (through this course or any other) to attend a hardware repair training session, he or she should insist on being involved in as much hands-on training as is possible.

Replacing a Power Supply.

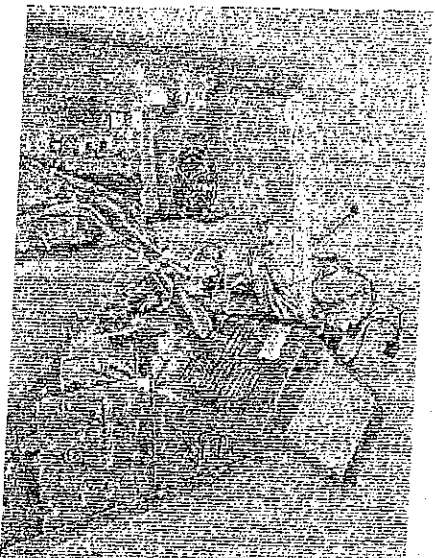
For desktop computers in developing countries, power supply failure is a common problem. This is due to an uneven or "dirty" voltage and power surges. Before replacing a power supply, a lab manager should make sure that it truly is dead by following the tip box. Then follow the steps below:

- (i) Unplug the power cord from the back of the computer case.
Open up the computer case. The method varies with the brand and model of computer, but often involves laying the case on its side and opening up the side panel. Some computer cases use screws and others have buttons to press that release one side panel. Find and unscrew all of the screws attaching the power supply to the back of the case. Usually there will be four screws, located just outside the cut out in the case (or along the mesh grid) where the power supply is found. Do not disconnect the various cables yet. Move the dead power supply as far out of the way as the cables will allow. You may be able to set the power supply on top of the case (see photo).

(ii) Install the new power supply and screw in all screws.

One by one, detach the cables from the old power supply to the various computer components, and replace them with the cables from the new power supply. Doing this one by one will help make sure that no components are missed. There will be power supply connectors for the motherboard and all drives. When removing the cables, keep an eye out for latches that need to be depressed to release the connector. All of the connectors should

Only fit in one direction, and with little force. If a connector does not attach easily, make sure that it is properly aligned before pushing harder.



Replacing a Hard Drive.

Replacing a faulty hard drive is not a difficult task, but a lab manager should have done some investigative work first to determine if the hard drive is faulty. This could include running the Chick disk utility, listening for unusual noises coming from the hard drive and also checking to make sure the cables to the hard drive are still fully connected. The computer may also present error codes during start up that indicate a problem with the hard drive. Before replacing a hard drive, it is worth trying to reset the BIOS to the default settings to see if this fixes the problem. To do this, enter the BIOS during computer start up and choose the default settings. Then save the new settings and exit. The onscreen instructions in the BIOS program will guide the computer lab manager through these steps. Once it has been determined that a hard drive problem is likely, the computer lab manager should follow the steps below to replace the hard drive with a known working hard drive.

Disconnect the computer case from electricity and then open the computer case (as described above).



(i) Locate the hard drive.

It will likely be found in a metal case or cage near the front of the computer case, below the other drives. It could be mounted directly on the cage or it could be mounted on rails that will slide out of the case. Unscrew the hard drive from the cage or slide the rails out and unscrew the hard drive from the rails.

(ii) Remove the long ribbon cable from hard drive.

There may have been a small amount of glue used to hold the ribbon in place that will also need to be removed. Then remove the smaller power cable. If it is difficult to remove, gently wiggle the connector back and forth along its long axis as you pull. Note the orientation of the connector (which end is the red wire on?) to make sure that the new hard drive is connected in the same manner.

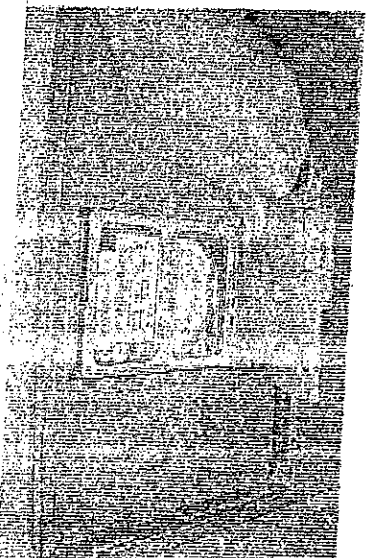
If there are two hard drives in a computer, one is called the master and the other the slave. On older hard drives, an electrical connection needs to be changed to indicate whether the hard drive is a master or a slave. If the hard drive being installed has not been used before, it will be correctly set as a master (or also on the acceptable "Cable Select" setting, used when the ribbon cable has two connectors -- one for a master and one for a slave). If the hard drive has been previously used as a slave, however, it will need to be changed to a master. To do this, look for a diagram on the hard drive that shows the pattern for connecting (jumping) two of the eight pins located between the two cable connectors on the front end of the hard drive. Then using needle nose pliers, pulls off the jumper that is connecting two of the pins and replace it to connect the proper pair.

(iii) Connect the cables to the new hard drive.

Then screw the hard drive back on to the rails and slide the rails into position (or screw the hard drive directly to the cage if there are no rails). Make sure that all of the other ribbons and cables are as they were found. Close the case and the repair is complete. Note that if any new noises or problems appear after the installation, it is possible that another device or cable was disrupted during the repair and needs to be reattached or reset.

Replacing a Hard Drive for a Laptop Computer.

Hard drives for many laptop computers can be replaced quite easily because the manufacturers have made the hard drives accessible from hatches on the underside of the computer. To replace the hard drive on a laptop with an accessible hard drive follow the steps below:



Remove the laptop battery before making this (or any other) laptop repair. This is important in order to avoid electricity from the battery flowing through the computer during repair, which could be damaging to computer components. The technique for removing the battery depends on the make and model of the laptop, but often involves unlocking the battery at one or two points (on the underside of the computer) and then sliding it out.

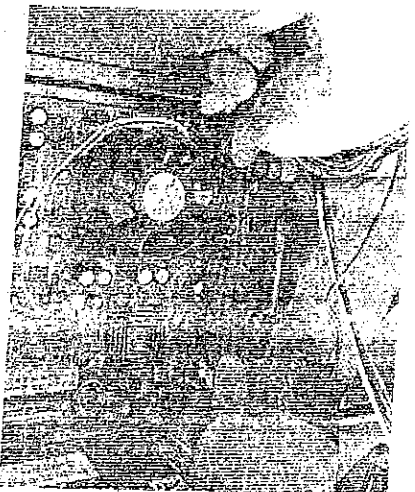
With the laptop resting upside down on a table, find the hard drive compartment and unscrew the screw or screws holding the lid in place. Remove the lid and examine how the hard drive is attached to the computer. Most likely you will see that the hard drive is attached to a cage or brackets that will slide out with the hard drive (see photo). Slide out the assembly containing the hard drive in its cage or brackets. You may have to push the assembly in or lift it up first in order to clear anything else used to hold the assembly in place.

Remove any screws attaching the laptop to its cage/brackets, remove the damaged hard drive, replace it with a functioning laptop hard drive and then retighten all screws. Slide the assembly back into the laptop and secure it as it was previously. Make sure that you push hard enough to reattach the hard drive to the power and data connectors. Close the hard drive compartment and replace all screws. Then reconnect the battery, turn the computer over and restart it.

Replacing RAM.

As a lab manager may want to add RAM to speed up a slow computer. RAM also can fail or come loose from its housing and need to be reseated. If a lab manager has decided that a computer needs new or additional RAM, it is important to make sure that the RAM used is compatible with the computer system. There are several online RAM sales companies that will identify the appropriate RAM if a user inputs the computer make and model. Two example companies are Corsair (www.corsairmemory.com) and Crucial (www.crucial.com). Once the RAM has been obtained, the lab manager should follow the steps below to replace the RAM. **Please Note:** RAM is very easily damaged by electrostatic charges, so it is very important for the user to be grounded before picking up RAM. Also RAM should only be handled by the edges.

Open the computer case as previously described. Find the RAM slots (long parallel connectors). Note which slots are occupied and if the slots are numbered (at the ends). If the slots are not numbered, the slots closest to the processor are probably the lower numbered slots and should be filled first.



Simultaneously push down and out on both retaining clips (on each end) of a RAM slot to remove the old RAM. Note the orientation of the RAM so that the replacement RAM can be put in facing the same direction. There should be some notches on the bottom edge of the RAM that can be used as a guide.

Gently place the new RAM in the socket and press down firmly on the top of the RAM at both ends (see photo). When the RAM module is correctly seated, the retaining clips will automatically close.

Replacing Laptop RAM.

Laptop computers have a compartment on the back side designed for accessing RAM. When the screws for this compartment are removed (but don't forget to remove the laptop battery first for safety reasons!), the RAM should be visible. For some laptops, however, only the slot for expansion (extra) RAM is visible. In these cases the rest of the RAM is can be accessed by removing the keyboard, a more difficult task (discussed below). Laptop RAM rests horizontally, with the slot on one end. To install RAM, hold the RAM by its edges, and bring it up towards its slot at about a 30 degree angle (see photo). Then gently push the RAM in while flattening out the angle.

